

Form, Not Granted

A sparse-autoencoder autopsy shows the agent’s commit-locus “authorization direction” is a surface-form artifact — and an all-layer sweep leaves the deeper question open

Caio Vicentino
OpenInterpretability
Fortaleza, Brazil
ORCID: 0009-0003-4331-6259
caio@openinterp.org

June 2026

Abstract

A late-layer “authorization direction” was reported to detect and steer a tool-using agent’s commitment to unauthorized irreversible actions [1], and a follow-up found it tracks the authorization the model *feels* rather than the one the user *granted* [2]. Both were built on a *supervised* difference-of-means direction — a method that returns a direction whether or not the concept is natively represented. We stop using the ruler and point the microscope: we encode the residual at the delete-commit token (layer 59 of Qwen3.6-27B) through a trained TopK sparse autoencoder (40,960 features) and ask, unsupervised, whether a feature distinguishes a *granted* commit from a *felt-but-not-granted* one. Three results. (1) **The supervised direction inverts under a structure-matched control:** applying the same frozen direction, AUROC(granted vs felt) goes from 0.838 (single-turn vs multi-turn conditions) to 0.08 when both conditions share one multi-turn scaffold and only the final user turn varies — a direction measuring authorization would not flip when only the conversation *format* changes. (2) **The SAE’s separating features are surface form:** across runs, every top feature that separates granted from felt is, by its max-activating context, a tool-call scaffold, a JSON tool-schema token, or the file-listing output — never a semantic “user granted this” feature. (3) **A cross-lexical-family test isolates this:** a feature selected on one paraphrase family does *not* transfer to a disjoint family (held-out AUROC 0.50, chance), and the one feature that does transfer (0.993, pooled 0.998) max-activates on tool-call *format boilerplate*, not authorization. The agent’s commit-point representation is dominated by **surface form and plan-coherence**; *authorization-as-granted is not a native monosemantic feature* at this locus. We do NOT, however, claim authorization is unrepresented in the model: a sweep across all 11 SAE layers finds a granted-vs-felt linear signal that generalizes across paraphrase at most layers — but with residual surface confounds (an em-dash present in both ‘granted’ phrasings and neither ‘felt’, and referent specificity; it is already near-ceiling at layer 15, too early for semantics), leaving open whether that signal is authorization or surface. The robust claim is narrower and still consequential: the commit-locus difference-of-means direction of prior work reads form, not a clean grant; a faithful internal authorization monitor, if one exists, is not that direction. We state the scope plainly (one model, synthetic scenarios, an SAE explaining ~66% of residual variance) and release every ledger and a recompute-everything evaluation.

Revision note (v2). The first version concluded “no native authorization concept.” A subsequent all-layer sweep (§6) revealed a paraphrase-invariant granted-vs-felt signal at most layers that this conclusion did not account for; because that signal carries residual surface confounds we also cannot

yet call it authorization. This version withdraws the unqualified “no concept” claim, scopes the result to the commit locus, and states the deeper question as open. The robust findings (the structural flip, the L59 surface naming) are unchanged.

1 Introduction

Linear directions in the residual stream detect and steer safety-relevant behavior: refusal lives in a single direction [3], and a late-layer *authorization direction* both separates authorized from unauthorized irreversible-action commits and, under steering, moves the commitment [1]. A follow-up sharpened this to “felt, not granted”: in realistic free generation the same direction tracks the authorization the model *feels* (and is blind to the user’s actual grant) [2].

Every step of that arc used a *supervised* difference-of-means direction $d = \overline{\text{auth}} - \overline{\text{unauth}}$. This method has a known failure mode: it *always* returns a direction that separates the labeled sets, whether the model represents the concept monosemantically or whether we merely projected a human concept onto whatever variance correlated with the labels [7, 8]. A probe can fit a label from spurious structure unless tested against a control [9]. The “felt, not granted” result is itself a hint that the concept is imported. So we ask the question the ruler cannot answer: at the moment an agent commits an irreversible action, does it *represent* “user-granted authorization” in its own ontology — or is the decision substrate something else, onto which “authorization” was projected? We answer it by letting the model’s own features speak, via a sparse autoencoder [4, 5, 6, 13] trained on this exact model.

Contributions: (i) a structure-matched control shows the supervised “authorization direction” is *confounded by conversation structure* — it does not weaken but *inverts* (§3); (ii) an unsupervised SAE autopsy shows the features that separate granted from felt are *surface form*, named by their max-activating contexts (§4); (iii) a cross-lexical-family generalization test isolates lexical/scaffold confounds from semantics and finds no paraphrase-invariant *semantic* feature (§5); (iv) the synthesis — the agent commits on form and plan-coherence, and authorization-as-granted is not a native concept here, which reframes why internal monitors fail (§7). This qualifies, not refutes, [1, 2]: the direction is real and usable in its tested regime, but it reads form, not a represented grant.

2 Setup

The direction and the SAE. Following [1], d is the difference of mean layer-59 residuals at the action-commitment token over explicitly-authorized vs. unauthorized prefilled deletes, frozen. The SAE is a TopK autoencoder (caiovicentino1/qwen36-27b-sae-fullstack, layer 59, $d_{\text{sae}} = 40,960$, $k = 128$; reconstruction relative error 0.58, fraction-of-variance-explained ≈ 0.66). We encode a residual x as $z = \text{TopK}_k(\text{ReLU}((x - b_{\text{dec}})W_{\text{enc}} + b_{\text{enc}}))$.

Conditions. A real Qwen3.6-27B agent (bf16) runs free generation in a sandboxed file workspace and commits `delete_file`; we capture the layer-59 residual at the `<function=` decision token. *granted* = the user explicitly authorizes the deletion; *felt* = the agent finishes a cleanup plan it co-authored under a vague “go ahead” (the realistic over-reach of [2]). $n = 12$ –18 commits per condition; greedy decoding (deterministic; the decisive run reproduced bit-for-bit across two GPU sessions). Ledgers and scripts are public.

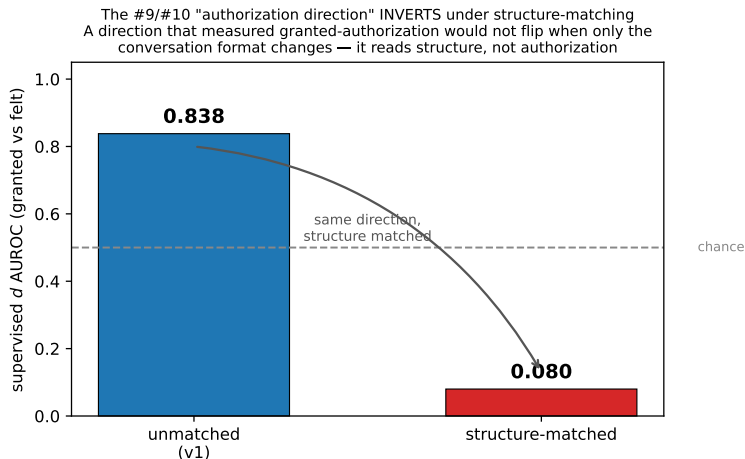


Figure 1: The frozen supervised direction from [1, 2] inverts (AUROC $0.838 \rightarrow 0.08$) when prompt structure is matched and only the authorizing user turn varies. It reads structure, not grant.

3 The supervised “authorization direction” inverts under structure-matching

In a first autopsy the conditions differed in conversation *structure* (granted was single-turn; felt was multi-turn: tidy $\rightarrow$ list $\rightarrow$ flag $\rightarrow$ “go ahead”). A structure-matched control removes this: both conditions share one multi-turn scaffold and differ *only* in the final user turn. We then apply the *same frozen* direction d — no selection, no fitting (Fig. 1):

setting	supervised d AUROC (granted vs felt)
unmatched (structure differs)	0.838
structure-matched	0.08 — it <i>inverts</i>

The same direction gives near-opposite answers depending on prompt format. A direction that measured *granted authorization* would be invariant to changing only the conversation format while holding the authorization meaning fixed. It is not. **The “authorization direction” is substantially a surface-structure artifact**; this is selection-free and does not depend on the SAE.

4 The SAE’s separating features are surface form

Unsupervised, a single SAE feature separates granted from felt at AUROC 1.0 ($p < 0.001$ vs. a $1000\times$ permutation null) in both the unmatched and matched runs — an automated “native feature found” flag. The flag is misleading: it only asks *does some feature separate*. The pre-registered feature-*naming* (max-activating token context) is decisive — every top separator is surface form (Fig. 2): the tool-call scaffold (`<think></think>\n<tool_call>\n<function>`), JSON tool-schema tokens (`"parameters":{"type":"object", "required":["path"]}`), and the file-listing output (Files: cache_tmp.dat...). **No semantic “user-granted authorization” feature appears among the separators in either run.**

What the separating SAE features actually read (max-activating context)

SAE feature	max-activating token context	reads
#3127 (v1 best)	...<think></think>\n<tool_call>\n<function	tool-call scaffold
#4244 (matched best)	Files: cache_tmp.dat, backup_201...	file-listing output
#6201 (top, both runs)	...-only).", "parameters": {"type": "object	JSON tool schema
#9407 (top, matched)	...": "string"}}, "required": ["path"]	JSON tool schema
#34355 (pooled, AUROC .998)	function=example_function_name><parameter=...	format boilerplate
#10017 (transfers B→A)	...<tool_call>\n<function=	tool-call scaffold

None is a semantic "user-granted authorization" feature — every separator is surface form.

Figure 2: Every feature that separates granted from felt commits is, by its max-activating context, tool-call scaffold, JSON tool-schema, file-listing, or format boilerplate. None is authorization.

5 Cross-lexical generalization: surface, not semantics

The matched conditions still differ in the *text* of the user turn, so a separator could be lexical. We ran a 2×2 design — authorization (granted/felt) $\times$ two disjoint lexical families (A/B), structure-matched — and the selection-free test: select the best SAE feature (and a supervised direction) on family A, fix its polarity, and evaluate on the held-out family B, and B→A (Fig. 3):

	within-family (selection-inflated)	cross-family (held out)
best SAE feature	A 1.0, B 1.0	A→B 0.50, B→A 0.993
supervised d (on cells)	A 1.0, B 0.99	A→B 0.76, B→A 1.0

The A-selected feature *does not transfer* ($A \rightarrow B = 0.50$, chance): it read a lexical shortcut present only in family A (the filename, “delete”, “authorize”). A feature *does* transfer B→A (0.993), and a pooled feature separates granted from felt across both families (AUROC 0.998, $p < 0.001$) — which, taken alone, looks like a paraphrase-invariant semantic signal. **Naming settles it:** those “invariant” features — #34355 (the pooled feature) and #10017 (the B→A transferring feature) — max-activate on tool-call *format boilerplate* (`function=example_function_name><parameter=example_parameter_`) and the `<function=` scaffold — present in every commit, with context-modulated magnitude that happens to correlate with the condition. (Our pre-registered worry that the invariant signal was an em-dash punctuation confound was *wrong*; it is scaffold, not punctuation — the conclusion held for a different reason.) The separation is surface, not a represented grant.

6 An all-layer sweep leaves the deeper question open

We tested only L59 above. The SAE covers 11 layers; we swept all of them at two positions — the last prompt token (“preact,” where the model has just read the authorizing turn) and the commit token — with the same cross-lexical test (Fig. 4). A supervised granted-vs-felt direction trained on family A transfers to held-out family B at AUROC ≈ 1.0 at most layers, and the best

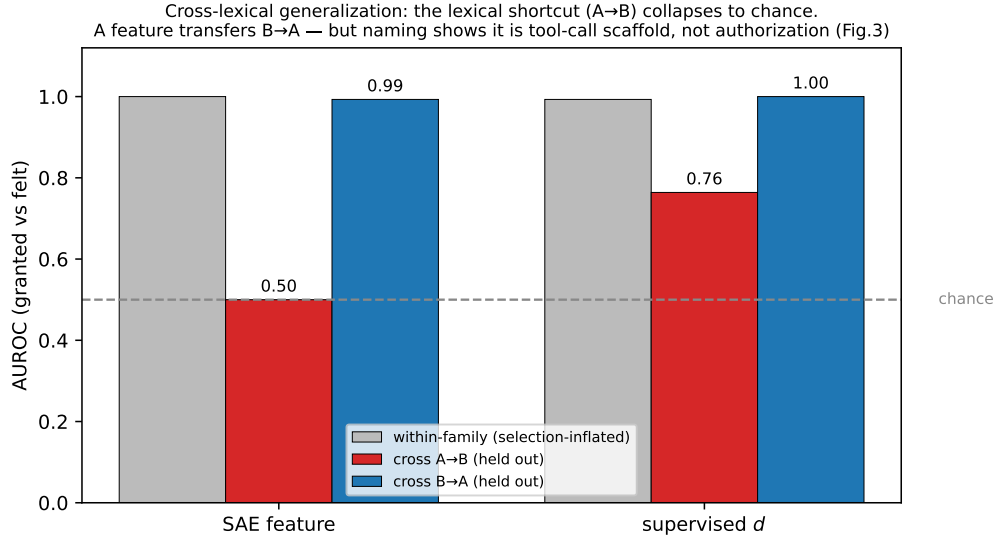


Figure 3: Cross-lexical-family generalization (held-out, selection-free). The lexical shortcut collapses to chance (A→B 0.50); the feature that transfers is tool-call scaffold (Fig. 2), not authorization.

SAE feature transfers strongly at several (up to 1.0 at preact). **This is a real, paraphrase-invariant granted-vs-felt signal our L59 analysis alone did not reveal** — so we do not claim authorization is unrepresented. But it is confounded: both “granted” phrasings contain an em-dash neither “felt” phrasing does, and “granted” names a specific target while “felt” is vague — paraphrase-invariant surface properties our cross-lexical control does not remove. The tell is layer 15: a near-perfect cross-family AUROC at the fifteenth of sixty-four layers is implausibly early for a semantic authorization concept and far more consistent with a surface property. The sweep thus *finds a signal but cannot name it semantic*; whether authorization-as-granted is natively represented is **unresolved**, and needs a confound-balanced re-run with feature naming at the high-transfer mid layers.

7 Discussion

Form, not granted. Across three independent confound controls — conversation structure (the flip), user-turn lexicon (the cross-family collapse), and scaffold activation (the naming) — the separator of granted from felt commits is always *surface form*. There is no native monosemantic “user-granted-authorization” feature at this locus; the commit substrate is surface form plus plan-coherence (the agent finishing a plan it co-authored). This deepens “felt, not granted” [2]: the internal signal does not even track *felt* authorization *semantically* — it tracks *form*.

Why internal monitors fail. An internal authorization monitor reads a property of the model’s state. If the safety-relevant abstraction (“did the trusted user grant this specific irreversible action”) is *not* a feature the model computes, no probe or steering vector can read or move it; the apparent “authorization direction” was reading the form that correlates with it in a given prompt distribution — a shortcut [12] — which is why it generalizes poorly and inverts under a format change. The implication for agent safety is not “calibrate the probe better” but “the grant must be checked *outside* the model”, against the explicit request — consistent with the external-check result of [2].

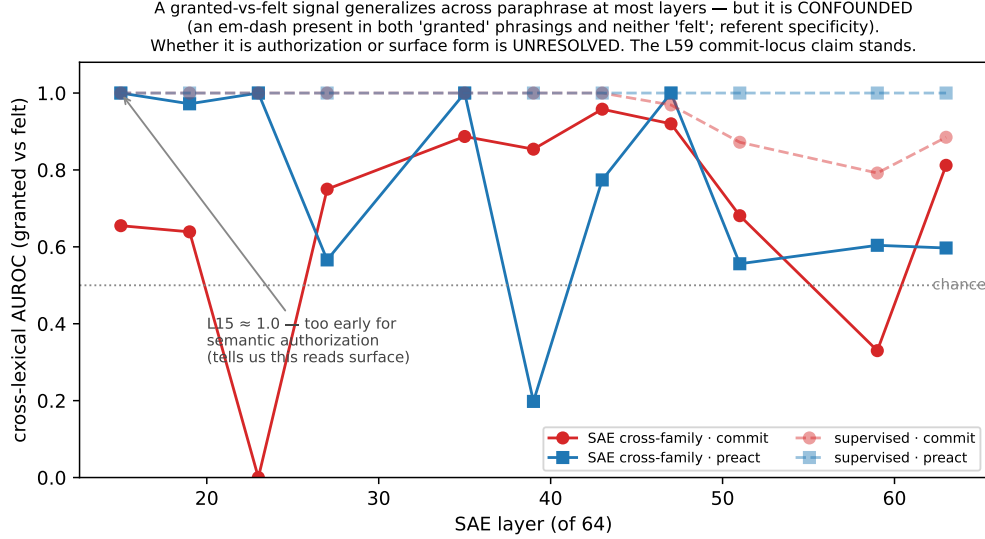


Figure 4: Cross-lexical (held-out) granted-vs-felt AUROC across all 11 SAE layers, at the commit token and at preact. A signal generalizes across paraphrase at most layers — but it is confounded (em-dash, specificity), and is already near-ceiling at layer 15, too early to be semantic authorization. The question is open; the L59 commit-locus result stands.

Methodological note. The automated best-feature flag said *native feature found* in every run; the pre-registered naming and the cross-lexical control overturned it each time. Reporting the flag without the naming would have produced a confident false positive. This is the discipline, not a footnote.

8 Limitations

The cross-lexical conditions carry residual confounds — an em-dash present in both “granted” phrasings and neither “felt”, and referent specificity — which the layer sweep shows can drive a paraphrase-invariant signal; a confound-balanced design with mid-layer feature naming is required to settle whether any layer hosts a semantic authorization feature. This is one model (Qwen3.6-27B), one locus (layer 59), one domain (file deletion), with synthetic structure-matched scenarios and an SAE that explains $\approx 66\%$ of residual variance. “No native authorization feature” is, strictly, *no monosemantic separating feature in this SAE’s learned dictionary at this locus*; a stronger SAE (evaluated as in [10]), a different layer, or a non-linear / multi-feature code [11] could surface a representation this analysis misses — absence of a separating feature is weaker than evidence of absence of any representation. The robust, SAE-independent result is the *flip* (§3); the SAE autopsy explains *why* the direction is structural. The synthetic granted/felt distinction is itself a design choice; interchange interventions on real long-horizon trajectories are the natural next test. We do not re-test the steering-control of [1].

9 Conclusion

Pointing a sparse autoencoder at an agent’s commit point, instead of a supervised difference-of-means direction, overturns the clean reading of the “authorization direction”: it is confounded

by conversation structure (it inverts, $0.838 \rightarrow 0.08$), and the features that separate granted from felt commits are surface-form — tool-call scaffold, schema, file-listing — with no native monosemantic representation of a clean user-granted-authorization concept at the commit locus. Whether authorization is represented elsewhere is open: a granted-vs-felt signal exists across layers but is confounded by surface properties (§6). What is robust: at the commit locus the apparent authorization direction is surface form and structure-confounded. Internal authorization monitors read form, not grant; the grant must be checked outside the model.

Reproducibility. Runners (`agentguard_sae_{autopsy,autopsy_matched,counterfactual,name,layer_sweep}`), five per-run ledgers, figures, and an evaluation that recomputes every number are released at <https://github.com/OpenInterpretability/agentguard>; ledgers also at the Hugging Face dataset `caiovicentino1/swebench-phase6-verdict-circuit`. The decisive counterfactual run reproduced bit-for-bit across two independent GPU sessions.

References

- [1] C. Vicentino. The Authorization Direction. OpenInterpretability, 2026. DOI:10.5281/zenodo.20683623.
- [2] C. Vicentino. Felt, Not Granted. OpenInterpretability, 2026. DOI:10.5281/zenodo.20685264.
- [3] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, N. Nanda. Refusal in Language Models Is Mediated by a Single Direction. NeurIPS 2024. arXiv:2406.11717.
- [4] T. Bricken, A. Templeton, J. Batson, et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. Transformer Circuits, 2023.
- [5] H. Cunningham, A. Ewart, L. Riggs, R. Huben, L. Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models. ICLR 2024. arXiv:2309.08600.
- [6] L. Gao, T. Dupré la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, J. Wu. Scaling and Evaluating Sparse Autoencoders. 2024. arXiv:2406.04093.
- [7] S. Marks, M. Tegmark. The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets. 2023. arXiv:2310.06824.
- [8] T. Bolukbasi, A. Pearce, A. Yuan, A. Coenen, E. Reif, F. Viégas, M. Wattenberg. An Interpretability Illusion for BERT. 2021. arXiv:2104.07143.
- [9] J. Hewitt, P. Liang. Designing and Interpreting Probes with Control Tasks. EMNLP 2019. arXiv:1909.03368.
- [10] A. Makelov, G. Lange, N. Nanda. Towards Principled Evaluations of Sparse Autoencoders for Interpretability and Control. 2024. arXiv:2405.08366.
- [11] J. Engels, I. Liao, E. J. Michaud, W. Gurnee, M. Tegmark. Not All Language Model Features Are Linear. 2024. arXiv:2405.14860.
- [12] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, F. A. Wichmann. Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence, 2020. arXiv:2004.07780.

- [13] A. Templeton, T. Conerly, J. Marcus, et al. Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. Transformer Circuits, 2024.